PRACTICE QUIZ — ANSWER KEY

40 multiple choice · 15 true/false · 3 matching sets — his stated format. Closed book, 6:35pm.

Part A · Multiple Choice

1. Blood is classified as a connective tissue because:

A) It has a matrix with cells suspended in it ◀ correct
B) It is produced inside bone
C) It contains collagen fibres
D) It connects organs to one another
The definition of connective tissue is a matrix + suspended cells — not solidity. Plasma is the matrix; the formed elements are the cells. Being produced in bone is true but irrelevant to the classification.

2. The normal pH range of blood is:

A) 7.25 – 7.35
B) 7.35 – 7.45 ◀ correct
C) 7.45 – 7.55
D) 7.00 – 7.20
7.35–7.45, slightly alkaline. Below 7.35 = acidic, above 7.45 = alkaline.

3. Blood viscosity is determined primarily by:

A) Plasma protein concentration
B) White blood cell count
C) The number of red blood cells ◀ correct
D) Water content of plasma
RBCs are 45% of blood volume; WBCs and platelets together are under 1%. The spun tube makes the argument visually.

4. After centrifugation, plasma represents approximately what fraction of blood volume?

A) 45%
B) 90%
C) 70%
D) 55% ◀ correct
Plasma 55%, RBCs 45%, buffy coat (WBCs + platelets) under 1%. Don't confuse 55% with the 90% figure — that's water as a fraction OF plasma.

5. The dominant protein in plasma is:

A) Albumin ◀ correct
B) Thrombin
C) Fibrinogen
D) Gamma globulin
Albumin, roughly 60% of plasma protein. This was the answer he asked the class to supply.

6. The primary function of albumin is to:

A) Transport oxygen
B) Maintain osmotic pressure ◀ correct
C) Act as an antibody
D) Form the clot mesh
Albumin holds water inside the vessel. Lose it and water moves into the tissue → edema.

7. Most plasma proteins are manufactured by the:

A) Red bone marrow
B) Kidneys
C) Liver ◀ correct
D) Spleen
The liver. This connects directly to why liver disease causes bleeding — clotting factors are proteins.

8. The swollen abdomen seen in Kwashiorkor is caused by:

A) Enlarged liver
B) Intestinal parasites
C) Excess fat deposition
D) Fluid leaving the circulation into the tissue ◀ correct
↓ dietary protein → ↓ albumin → ↓ osmotic pressure → water leaves the vessels. In his words: “the big belly is not fat; it's just water — you can even see the ribs.”

9. Mature erythrocytes lack all of the following EXCEPT:

A) Hemoglobin ◀ correct
B) A nucleus
C) Mitochondria
D) Ribosomes
Hemoglobin fills 97% of the cell. Everything else is discarded to make room for it.

10. The biconcave shape of the RBC is advantageous because:

A) It makes the cell heavier than plasma
B) No interior point lies far from the membrane, speeding gas exchange ◀ correct
C) It protects the nucleus
D) It allows the cell to store more iron
His sphere analogy: in a ball the core sits far from the surface. In a disc, nothing does. (And there is no nucleus to protect.)

11. Red blood cells generate ATP anaerobically. The functional consequence is:

A) They can survive without glucose
B) They produce ATP more efficiently than other cells
C) They do not consume the oxygen they carry ◀ correct
D) They divide more rapidly
Lacking mitochondria does two things: frees space for hemoglobin, and means the cell never burns its own cargo. 100% of the O_2 is delivered.

12. Oxygen binds to which part of the hemoglobin molecule?

A) The globin protein chains
B) The ribosomes
C) The cell membrane
D) The iron atom at the centre of each heme ◀ correct
Fe^{2+} at the centre of heme. This is why iron deficiency causes fatigue — no iron, no O_2 carriage.

13. One red blood cell can carry approximately how many oxygen molecules?

A) 1 billion ◀ correct
B) 250 million
C) 4
D) 250 thousand
4 O_2 per hemoglobin × 250 million hemoglobin per RBC = 1 billion. Reconstruct it rather than memorising it. "We are all billionaires."

14. The average lifespan of an erythrocyte is:

A) 30–40 days
B) 100–120 days ◀ correct
C) 1 year
D) 60–80 days
100–120 days. With no nucleus it cannot repair itself, so it wears out.

15. Aged and damaged red blood cells are removed primarily by the:

A) Kidneys
B) Liver
C) Spleen ◀ correct
D) Red bone marrow
The spleen is the main site; the liver contributes to a smaller extent.

16. Erythropoietin is secreted by the ______ in response to ______.

A) Red bone marrow … blood loss
B) Kidneys … hyperoxia
C) Liver … hypoxia
D) Kidneys … hypoxia ◀ correct
Kidneys (liver to a small extent), triggered by hypoxia. Kidney → EPO → red marrow → more RBCs → O_2 restored → stimulus switches off.

17. A reticulocyte is best described as:

A) An immature RBC that has ejected its nucleus but retains ribosomes ◀ correct
B) A white blood cell precursor
C) An RBC that has lost its hemoglobin
D) A fragment of a megakaryocyte
Nucleus gone, ribosomes still present. Matures in the bloodstream in about 2 days. Normal = 1–2% of RBCs.

18. A patient's reticulocyte count is 10%. This most likely indicates:

A) A primary bone marrow disease
B) The bone marrow is compensating for a problem elsewhere ◀ correct
C) Normal healthy variation
D) Vitamin B12 toxicity
A student asked exactly this. It is a COMPENSATORY MECHANISM, not a disease — the marrow is responding correctly to bleeding, hemolysis, or newly available iron. The pathology is somewhere else.

19. Hematopoiesis occurs in:

A) Yellow bone marrow
B) The liver in adults
C) Red bone marrow ◀ correct
D) The spleen
Red marrow only. Yellow marrow is fat storage — and red converts to yellow with age.

20. Which is NOT a typical site of adult hematopoiesis?

A) Proximal femur
B) Pelvic girdle
C) Vertebrae
D) Distal tibia ◀ correct
Axial skeleton, pelvis, and the PROXIMAL ends of long bones. Distal long bone = yellow marrow.

21. Which nutrient sits at the centre of the heme group?

A) Iron ◀ correct
B) Folate
C) Calcium
D) Vitamin B12
Iron. B12 and folate matter for DNA synthesis during division, but iron is the O_2-binding site itself.

22. The most abundant white blood cell is the:

A) Monocyte
B) Neutrophil ◀ correct
C) Lymphocyte
D) Basophil
Neutrophils 50–70%, lymphocytes second at 25–33%. He said not to memorise exact percentages but DO know the rank order: Never Let Monkeys Eat Bananas.

23. An elevated eosinophil count most strongly suggests:

A) Bone marrow failure
B) Viral infection
C) Parasitic infection or allergy ◀ correct
D) Bacterial infection
He flagged this explicitly as licensing-exam material. Eosinophilia → parasites, allergies, asthma. Think eczema, allergic rhinitis, seasonal allergy.

24. Which white blood cell develops into a macrophage in the tissues?

A) Neutrophil
B) Eosinophil
C) Basophil
D) Monocyte ◀ correct
Monocytes are the precursors of macrophages.

25. Histamine is released primarily by:

A) Basophils ◀ correct
B) Erythrocytes
C) Monocytes
D) Neutrophils
Basophils — the rarest WBC. They also carry heparin, an anticoagulant.

26. The ability of a white blood cell to squeeze through a vessel wall into tissue is called:

A) Phagocytosis
B) Diapedesis ◀ correct
C) Hemostasis
D) Chemotaxis
Diapedesis = "leaping across." Chemotaxis is following the chemical trail once outside. RBCs cannot do this; they stay in circulation.

27. Leukopenia is most commonly induced by:

A) High altitude
B) Stress
C) Chemotherapy ◀ correct
D) Bacterial infection
Chemotherapy suppresses the bone marrow — a student supplied this answer in class. It recovers once chemo stops.

28. Platelets are formed from:

A) Reticulocytes
B) Monocytes
C) Lymphoblasts

D) Megakaryocytes ◀ correct

They are cytoplasmic FRAGMENTS of megakaryocytes — not cells in their own right.

29. The hormone that regulates platelet production is:

A) Thrombopoietin ◀ correct
B) Erythropoietin
C) Nitric oxide
D) Thyroxine

Thrombo (platelet) + poietin (production). The naming is systematic — compare erythropoietin.

30. Platelets are normally kept inactive in circulation by:

A) Heparin secreted by basophils
B) Nitric oxide secreted by endothelial cells ◀ correct
C) Albumin
D) Low blood calcium

Nitric oxide from the vessel lining. Activation is not a switch turning on — it is this suppression being interrupted when collagen is exposed.

31. Place the steps of hemostasis in the correct order:

A) Vascular spasm → fibrin mesh → platelet plug
B) Fibrin mesh → vascular spasm → platelet plug
C) Vascular spasm → platelet plug → fibrin mesh ◀ correct
D) Platelet plug → vascular spasm → fibrin mesh

Spasm, plug, mesh. He called vascular spasm “probably the most important factor” because it acts immediately — but it is never sufficient alone.

32. Which conversion produces the insoluble threads that seal a wound?

A) Albumin → globulin
B) Prothrombin → thrombin
C) Heme → bilirubin
D) Fibrinogen → fibrin ◀ correct

Fibrinogen (soluble, already in plasma) → fibrin (insoluble mesh). The -ogen ending marks the precursor. Prothrombin → thrombin is the enzyme step that drives it.

33. A clot that forms and remains in an unbroken vessel is called a(n):

A) Thrombus ◀ correct
B) Embolus
C) Infarct
D) Hematoma

Thrombus = stationary, no trauma. It becomes an embolus the moment it travels. DVT is the classic thrombus.

34. A patient with a known DVT is at greatest risk of pulmonary embolism when they:

A) Sleep
B) Begin walking ◀ correct
C) Remain seated
D) Elevate the legs

The cruel irony he pointed out: a stationary DVT is manageable. The danger comes when the clot breaks loose — often as the patient starts moving.

35. Low-dose aspirin prevents clots by:

A) Inhibiting vitamin K
B) Dissolving existing clots
C) Blocking platelet aggregation ◀ correct
D) Destroying platelets

He was explicit: “not destroying platelets, just the ability for aggregation kind of reduces.” The platelet count stays normal.

36. Warfarin (Coumadin) works by:

A) Inhibiting thrombin directly
B) Blocking platelet aggregation
C) Increasing nitric oxide
D) Interfering with vitamin K ◀ correct

Vitamin K. Clotting is a cascade — every link must be present, so removing one collapses the chain. Heparin is the injectable that inhibits thrombin; aspirin is the anti-aggregant.

37. Newborns receive a vitamin K injection at birth because:

A) Their gut has not yet been colonised by the bacteria that produce it ◀ correct
B) It prevents jaundice
C) Breast milk lacks vitamin K entirely
D) Their liver cannot process vitamin K

Vitamin K is made by gut bacteria, and a newborn's gut is not yet colonised. The injection is prophylaxis so that any bleed doesn't become fatal.

38. Severe liver disease causes a bleeding tendency because the liver:

A) Filters old red blood cells
B) Manufactures clotting factors, which are proteins ◀ correct
C) Stores platelets
D) Produces erythropoietin
He asked this deliberately and a student guessed erythropoietin — wrong, that's the kidney. Clotting factors are proteins → the liver makes proteins → liver fails → no fibrinogen → bleeding.

39. Secondary polycythemia at high altitude is best described as:

A) An autoimmune disease
B) A form of anemia
C) A compensatory physiological response ◀ correct
D) A bone marrow cancer
Thin air → hypoxia → EPO → more RBCs. It resolves at sea level. Polycythemia VERA is the cancer — don't confuse the two.

40. After withdrawing an acupuncture needle that produces a drop of blood, you should:

A) Leave the point uncovered
B) Press and gently massage the point
C) Apply firm massage for 30 seconds
D) Press and hold with a cotton ball ◀ correct
Press and hold. The plug and mesh form within the first minute and are still fragile — massage tears them. The bruise may not appear until the next day.

Part B · True / False

1. Blood thinners work by reducing the number of red blood cells. ◀ FALSE

No blood thinner touches RBCs. Aspirin blocks aggregation; heparin inhibits thrombin; warfarin blocks vitamin K. The RBC count is unaffected.

2. Aspirin reduces the ability of platelets to aggregate without destroying them. ◀ TRUE

His exact framing: "not destroying platelets, just the ability for aggregation kind of reduces."

3. Atherosclerosis is caused primarily by plaque buildup rather than by blood viscosity. ◀ TRUE

A student asked whether thick blood causes it. Plaque (lipid) is the mechanism; viscosity affects how blood moves — a separate issue.

4. An elevated reticulocyte count is a compensatory response rather than a disease. ◀ TRUE

He was firm on this. The marrow is responding correctly; the pathology is elsewhere — bleeding, hemolysis, or iron repletion.

5. Thyroid hormone directly regulates erythropoiesis. ◀ FALSE

Thyroid sets cellular metabolic rate. Any connection to RBC production is INDIRECT — he was asked and said so.

6. Neutrophils, lymphocytes, monocytes, eosinophils and basophils are all granulocytes. ◀ FALSE

He corrected a student on exactly this. Lymphocytes and monocytes are AGRANULOCYTES.

7. Natural killer cells contain granules that are simply not visible under light microscopy. ◀ TRUE

True — which is why the granulocyte/agranulocyte split is a historical artifact of the optics available at the time, not a statement about biology.

8. A white blood cell count alone can be used to indicate sepsis. ◀ TRUE

A student asked whether you must wait for culture. His answer: no. Sepsis kills in hours via blood pressure collapse, so clinicians start broad-spectrum antibiotics immediately.

9. A stressful day can raise a person's white blood cell count. ◀ TRUE

He said he asks about the day of the draw when a count sits high. A raised WBC is not automatically infection.

10. Plasma proteins serve as an important fuel source for blood cells. ◀ FALSE

He lingered on this. Blood cells sit in a protein-rich fluid and never touch it — they use glucose, fatty acids and oxygen instead.

11. Increasing nitric oxide as much as possible improves platelet function. ◀ FALSE

A student asked this. His answer: the NORMAL amount. "Nothing to say the higher is the better or the lower is the better."

12. The red blood cell makes ATP anaerobically and does not consume the oxygen it carries. ◀ TRUE

No mitochondria. Lacking them both frees space for hemoglobin and means the cell never burns its cargo.

13. A cell committed to the myeloid pathway can later become a lymphocyte. ◀ FALSE

"Once committed to a pathway, they cannot change." Myeloid and lymphoid lines are separate.

14. Lymph, like blood, is classified as a connective tissue. ◀ TRUE

He said "blood and also lymph." Both have a matrix with suspended cells.

15. Platelets are true cells with a nucleus and full organelles. ◀ FALSE

Platelets are cytoplasmic FRAGMENTS. White blood cells are the only true cells among the formed elements.

16. Vascular spasm alone is sufficient to stop bleeding from a damaged vessel. ◀ FALSE

It is the most important single factor and acts fastest, but a small leak persists — which is exactly why steps 2 and 3 exist.

Part C · Matching

1. Match the hormone to its source and target.

Term	Match
Erythropoietin	**A** — Kidney → red bone marrow; triggered by hypoxia
Thrombopoietin	**B** — Liver/kidney → megakaryocytes; regulates platelet output

2. Match each drug to its mechanism.

Term	Match
Aspirin	**A** — Blocks platelet aggregation
Heparin	**B** — Inhibits thrombin; injectable, used in hospital
Warfarin	**C** — Interferes with vitamin K; oral, used after discharge

3. Match each anemia to its cause family.

Term	Match
Iron deficiency anemia	**B** — Production fails

Term	Match
Aplastic anemia	**C** — Production fails — marrow stops making all lines
Sickle cell anemia	**D** — Destruction / defect
Heavy menstruation	**A** — Blood loss